

NS-WSG Boards - Parts Shopping List (5 card builds)

Quantities populate ALL 5 cards + riser headers, with spares. BOM detail: hardware/ORDER.md

Boards ordered from PCBWay 2026-08-31 (10x riser, 5x card). (c) 2026 David Troy, in memory of Stephen Troy.

Qty	Part	Package / detail	Refs	Source / note
SEMICONDUCTORS & MODULES				
☐	5	Raspberry Pi Pico WH	RP2040 + WiFi, headers pre-soldered	module official reseller; ~\$7 ea
☐	10	SN74LVC245AN	DIP-20, octal bus xcvr	U1, U2 TI; DigiKey/Mouser
☐	10	SN74HCT541N	DIP-20, octal buffer	U3, U4 TI; DigiKey/Mouser
☐	5	SN74HCT138N	DIP-16, 3-to-8 decoder	U5 TI; DigiKey/Mouser
☐	6	PT8211-S	SOP-8, 16-bit stereo DAC	U6B LCSC/Alibaba; +1 spare (tiny, cheap)
SOCKETS & HEADERS				
☐	20	DIP-20 socket	0.3 in	U1-U4
☐	5	DIP-16 socket	0.3 in	U5
☐	5	DIP-8 socket	0.3 in	U6 alt only if using DIP PT8211/adaptor
☐	10	Female header 1x20, 0.1 in	Pico socket	module or cut from longer strips
☐	2	Male header strip 0.1 in	40-pin breakaway, x5 strips ea	J2 + risers each riser takes 2x15; J2 takes 2 pins
PASSIVES (qty covers 5 builds + spares)				
☐	40	0.1 uF ceramic	50V, 0.1 in lead spacing	C1-C6, C10 7 per board
☐	18	10 uF electrolytic	16V+ radial	C7, C8, C13 3 per board
☐	18	47 uF electrolytic	16V+ radial	C9, C11, C12 3 per board; audio coupling + VDD bulk
☐	8	10 ohm 1/4W	axial	R6
☐	8	1 k 1/4W	axial	R4
☐	8	10 k 1/4W	axial	R5
CONNECTORS & MISC				
☐	5	CUI SJ1-3523N	3.5 mm TRS jack, THT horizontal	J1 DigiKey/Mouser (CUI Devices)
☐	1	3.5 mm stereo cable or powered speakers		test audio
☐	1	Hookup wire 24-26 AWG	small spool	J2 mix-in
OPTIONAL / NICE TO HAVE				
☐	2	PAM8302 amp module	mono class-D breakout	drives a bare 8-ohm speaker from J1
☐	2	SOP-8 to DIP-8 adapter		U6 alt only if SOP-in-socket preferred
☐	1	Micro-USB cable	data-capable	Pico flashing if none on hand

Everything is through-hole except the PT8211-S (SOP-8, 1.27 mm pitch - easy drag solder).

TI DIP logic: verify 'N' suffix (PDIP) when ordering. Jack: CUI Devices SJ1-3523N exactly - the footprint matches its datasheet.